Exhaustivity in Hungarian Focus Constructions: A Visual-World Paradigm Study

Gábor PARTI, Yu-Yin HSU

The Hong Kong Polytechnic University

gabor.parti@connect.polyu.hk, yu-yin.hsu@polyu.edu.hk

Department of Language Science and Technology 語言科學及技術學系

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1. Introduction

Hungarian has syntactically marked focus with a **pre-verbal focus** (PVF) construction (1) which has been considered to entail exhaustivity (É.Kiss, 1998). This has since been challenged, and we can force *exhaustive* (2) and *contrastive* (3) readings as well:

1) *Anna* ***kávét*** *ivott.*
Anna coffee-ACC drink-PST-3SG
"Anna drank **coffee**."

a) [...and nothing else]
b) [...and not something else]

2) *Anna csak* ***kávét*** *ivott.*
"Anna drank only coffee."

3) *Anna többek között* ***kávét*** *ivott.*
"Anna drank, among other things, coffee."

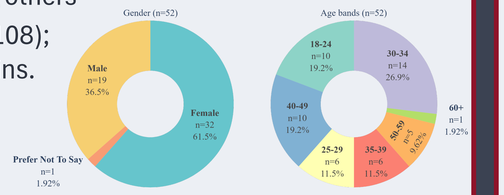
Research Questions:

- Is the pre-verbal focus interpretation exhaustive or contrastive by default?
- Is a contrastive reading available in a PVF construction?

2. Experiment

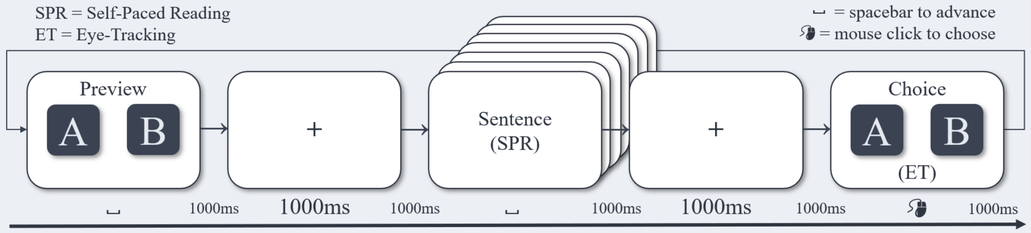
Sentence–Picture Verification w/ **Self-Paced Reading (SPR)** & **Eye-Tracking (ET)**

- Binary choice on visual stimuli after sentence using `PCIbex` (Schwarz & Zehr, 2021)
- 54 PVF sentence stimuli (18 items × 3 conditions):
  - **a) Unmodified** focus
  - **b) Exhaustive** focus w/ *csak* 'only'
  - **c) Contrastive** focus w/ *többek között* 'among others'
- Randomized; counterbalanced; + 54 fillers (Σ108);
- 40 trials (+3 practice) / experiment, ~25-30 mins.
- 55 participants (32 female; mean age=35)
- RTs collected during SPR and choice
- ET collected during choice



SPR = Self-Paced Reading
ET = Eye-Tracking

␣ = spacebar to advance
🖱 = mouse click to choose



3. Analysis

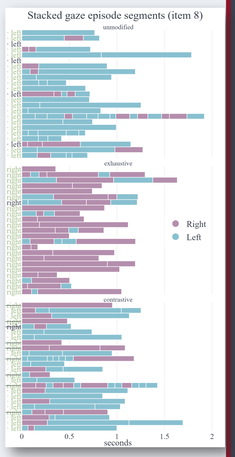
Eye-tracking (ET) data:

Cleaning & classification:

- 60ms threshold,
- 50ms neutral bridge gap
- 2 AOIs (left & right image)
- samples → gaze episodes → dwell

Metrics:

- Total dwell time (duration of all gaze episodes)
- Reading times (sentence)
- Reading times (regions)
- Reaction times (decision)



Modeling:

linear mixed-effects models (LME) with `lmer` in R:

*'y ~ Condition * Choice + (1 | Participant) + (1 | Item)*

- *'y = log(1 + y), for each metric*

Inference:

- Type-III Wald χ2 tests
- Tukey-adjusted pairwise comparisons (with `emmeans`)

4. Results

Behavioral results:

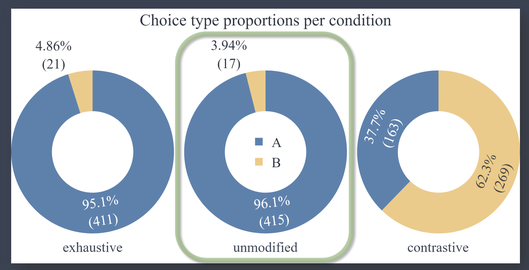
- Participants overwhelmingly interpreted unmodified PVF sentences like exhaustive *only*-sentences, choosing single-object images (A).
- In the contrastive condition multi-object ones (B) were preferred, implying a contrastive reading, while exhaustivity was still *available*.

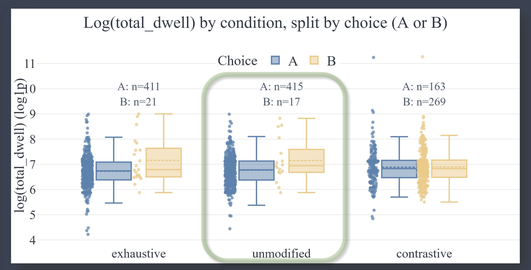
Reading times:

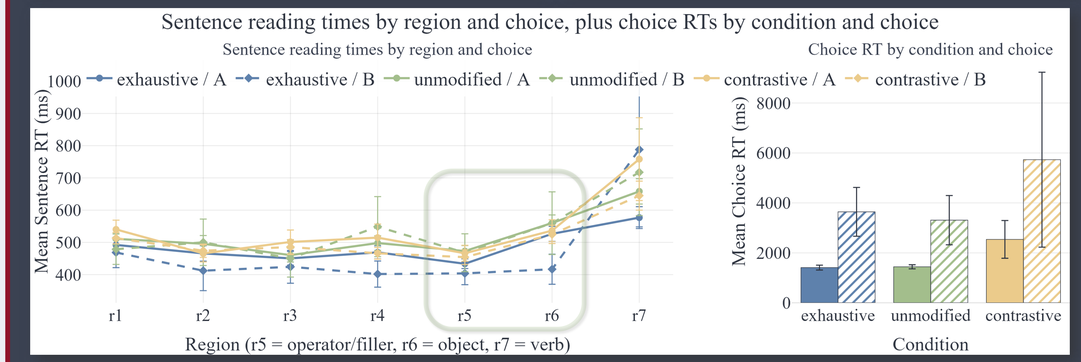
- Sentences were read faster in the exhaustive condition than both unmodified ($p<.0001$) and contrastive sentences ($p<.05$).
- Decision-making was quicker in the exhaustive condition than in contrastive ($p=.005$) regardless of the type, but not signif. diff. from the unmodified condition ($p=.140$). Decisions always took longer for multi-object choices ($p=.001$), suggesting that additional effort is needed.

Eye Tracking results:

- For total dwell times, significant effects were found for Condition ($\chi^2=12.531$, $p=.0019$), Choice ($\chi^2=11.845$, $p=.0006$), and their interaction ($\chi^2=12.292$, $p=.0021$). After reading exhaustive sentences, participants took significantly less time to choose A compared to B ($\beta=-.4443$, $p=.0081$), whereas they spent significantly more time to select A after reading a contrastive sentence than an exhaustive sentence ($\beta=.1876$, $p=.0057$). No significant difference was observed between selecting A or B for the unmodified PVF sentences ($\beta=-.2390$, $p=.5390$), implying that participants *might consider both representations equally plausible* when making their choices.







5. Conclusion

- Our results suggest that the PVF construction independently does **not** encode exhaustivity and it is compatible with both exhaustive and contrastive focus interpretations.
- It does, however, yield a strong **preference** towards exhaustive interpretations, but a contrastive reading is still available. We can **manipulate** this by modifying the PVF with specific phrases, such as 'among others', where the exhaustivity bias is weakened but persisting.
- In terms of the exhaustive vs. contrastive controversy in the literature, our results **support** the view that PVF constructions are not solely exhaustive (Onea & Beaver, 2009; Balogh 2013; Gerőcs et al. 2014; Káldi et al. 2016) through online sentence comprehension.
- Overall, the interpretation of PVF can be ambiguous, exhaustive choices were fast, while committing to a contrastive reading required more time and effort.

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